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Trek op meetkundige manier de vierkanstwortel in $\mathbb{C}$:

$$z^2 = -i$$

$$\begin{cases} r = \sqrt{0^2 + (-1)^2} \\ \tan \theta = -\frac{\pi}{2} \end{cases}$$

$$n = 2$$

$$\begin{aligned} \bullet \quad w_0 &= \operatorname{cis} \left(-\frac{\pi}{4} \right) \\ &= -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \end{aligned}$$

$$\begin{aligned} \bullet \quad w_1 &= \operatorname{cis} \left(\frac{-\frac{\pi}{2} + 2\pi}{2} \right) \\ &= \operatorname{cis} \left(\frac{3\pi}{2} \right) \\ &= \operatorname{cis} \left(\frac{3\pi}{4} \right) \\ &= \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \end{aligned}$$

References